The Lorentzian DeWitt Metric and the Pati-Salam Gauge Group

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March 5, 2026

Abstract

We study the DeWitt supermetric on the space of Lorentzian metrics over a four-dimensional spacetime manifold X. While the DeWitt metric evaluated on a Riemannian background is well known to have signature $(9, 1)$ on the ten-dimensional fibre of symmetric two-tensors, we show that on a *Lorentzian* background $g = \mathrm{diag}(-1, 1, 1, 1)$ the signature changes to $(6, 4)$. The normal bundle of a metric section $g\colon X \to Y^{14} = \mathrm{Met}(X)$ therefore carries structure group $\mathrm{SO}(6, 4)$, whose maximal compact subgroup is $\mathrm{SO}(6) \times \mathrm{SO}(4) \cong \mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$—the Pati-Salam gauge group. The Dynkin indices of $\mathrm{SU}(4)$, $\mathrm{SU}(2)_L$, and $\mathrm{SU}(2)_R$ in the fundamental representation of $\mathrm{SO}(6, 4)$ are all equal to one, giving the tree-level prediction $g_4 = g_L = g_R$ and hence $\sin^2\theta_W = 3/8$ at the unification scale, which runs to ≈ 0.231 at M_Z. We further identify the four negative-norm DeWitt modes as the $(1, 2, 2)$ Pati-Salam bidoublet—the Higgs sector—and show that the tree-level scalar potential is flat, realising a Gauge-Higgs Unification mechanism with a computable quartic coupling $\lambda(M_{\mathrm{PS}}) = g^2/4$. The Gauss equation for the embedding $g\colon X \hookrightarrow Y$ yields the Einstein-Hilbert, Yang-Mills, and torsion terms with the correct relative signs.

Keywords: DeWitt supermetric, Pati-Salam model, gauge-gravity unification, metric bundle, Gauge-Higgs Unification

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1 Introduction

The DeWitt supermetric [1] on the space of Riemannian metrics over a manifold plays a central role in canonical quantum gravity. Given a d-dimensional manifold X with metric g, the space $\mathrm{Met}(X)$ of all metrics on X is an infinite-dimensional manifold whose tangent space at g consists of symmetric two-tensors $h_{\mu\nu}$. DeWitt equipped this tangent space with the ultralocal metric

$$G^{\mathrm{DW}}(h,k) = \int_X \left(g^{\mu\rho}g^{\nu\sigma}h_{\mu\nu}k_{\rho\sigma} - \tfrac{1}{2}\,(g^{\mu\nu}h_{\mu\nu})(g^{\rho\sigma}k_{\rho\sigma})\right)\sqrt{|g|}\,d^dx\,. \tag{1}$$

At each point $x \in X$, the pointwise inner product on $S^2(T^*_xX)$ is

$$\mathcal{G}(h,k)\big|_x = g^{\mu\rho}g^{\nu\sigma}h_{\mu\nu}k_{\rho\sigma} - \tfrac{1}{2}\,\mathrm{tr}_g(h)\,\mathrm{tr}_g(k)\,, \tag{2}$$

where $\mathrm{tr}_g(h) = g^{\mu\nu}h_{\mu\nu}$.

For a Riemannian background ($g = \delta_{\mu\nu}$, signature $(+,+,+,+)$), the pointwise metric (2) on the $d(d{+}1)/2$-dimensional space of symmetric tensors has signature $(d(d{+}1)/2{-}1,1)$. In dimension $d = 4$ this gives $(9,1)$: nine positive directions and one negative (the conformal mode), a fact well known in the literature [2–4].

In this paper, we compute the signature of the pointwise DeWitt metric for a *Lorentzian* background metric $g = \mathrm{diag}(-1,1,1,1)$ and find that it changes to $(6,4)$. That the supermetric signature depends on the background signature was noted by Schmidt [5], whose general formula yields the same result; the gauge-theoretic consequences of this signature change, developed here, appear to be new.

Consider the bundle $Y = \mathrm{Met}(X)$ as a fibre bundle over X with projection $\pi\colon Y \to X$ sending each metric to its base point. A choice of metric $g\colon X \to Y$ is a section of this bundle. The total space Y is fourteen-dimensional: four base dimensions plus a ten-dimensional fibre $F = \mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(3,1)$. The DeWitt metric endows F with a pseudo-Riemannian metric of signature $(6,4)$.

The normal bundle N_g of the section $g(X) \hookrightarrow Y$ has fibre $\mathbb{R}^{6,4}$ and structure group $\mathrm{SO}(6,4)$. For a physical (compact) gauge theory, we pass to the maximal compact subgroup:

$$\mathrm{SO}(6,4) \;\supset\; \mathrm{SO}(6)\times\mathrm{SO}(4) \;\cong\; \mathrm{SU}(4)\times\mathrm{SU}(2)_L\times\mathrm{SU}(2)_R\,, \tag{3}$$

which is precisely the *Pati-Salam gauge group* [6].

This provides a purely geometric derivation of the Pati-Salam group from four-dimensional Lorentzian spacetime, with no free parameters in the gauge sector. The remainder of the paper establishes this result rigorously and explores its consequences.

Outline. Section 2 reviews the DeWitt metric and fixes conventions. Section 3 proves the $(6,4)$ signature for Lorentzian backgrounds and identifies the positive and negative eigenspaces physically. Section 4 derives the Pati-Salam group from the normal bundle. Section 5 computes the gauge coupling ratios and the Weinberg angle prediction. Section 6 identifies the Higgs sector as the $(1,2,2)$ bidoublet from the negative-norm modes and establishes tree-level flatness. Section 7 shows how the Gauss equation yields the correct gravitational and gauge field terms. Section 8 discusses open problems and relation to prior work.

Relation to prior work. Schmidt [5] derived a general formula for the supermetric signature as a function of background signature, from which the $(6,4)$ result follows; he did not pursue gauge-theoretic consequences. The idea of using the fourteen-dimensional metric bundle for unification appears in Weinstein's "Geometric Unity" programme [7,8], which also identifies the Pati-Salam group via $\mathrm{Spin}(6,4)$. Our approach differs in several respects: we use the *standard* DeWitt metric rather than a chimeric metric, we derive the signature $(6,4)$ from an explicit computation rather than an anthropic selection, and we employ the submanifold Gauss equation rather than a novel "shiab" operator. The present series further develops the fermion content (Paper II), gauge dynamics (Paper III), anomaly cancellation (Paper IV), and three-generation problem (Paper V)—consequences not addressed in the Geometric Unity framework.

2 The DeWitt metric

Let (X^d, g) be a d-dimensional pseudo-Riemannian manifold. The tangent space to $\mathrm{Met}(X)$ at g is the space $\Gamma(S^2T^*X)$ of symmetric two-tensor fields. At each point $x \in X$, we work in the fibre $V = S^2(T^*_xX)$, a vector space of dimension $n = d(d{+}1)/2$.

Definition 2.1 (DeWitt supermetric)**.** For $h, k \in S^2(T^*_xX)$, the pointwise DeWitt metric is

$$\mathcal{G}(h,k) = G^{\mu\nu\rho\sigma}\, h_{\mu\nu}\, k_{\rho\sigma}\,, \tag{4}$$

where

$$G^{\mu\nu\rho\sigma} = g^{\mu(\rho}g^{\sigma)\nu} - \tfrac{1}{2}\, g^{\mu\nu}g^{\rho\sigma}\,. \tag{5}$$

The more general one-parameter family of DeWitt metrics replaces the coefficient $-1/2$ with $-\lambda$ [2]. We work exclusively with the canonical choice $\lambda = 1/2$, which is distinguished by being the unique ultralocal metric on $S^2(T^*X)$ invariant under the full diffeomorphism group and compatible with the constraint structure of general relativity [1,2].

Remark 2.2. Our conventions: Greek indices $\mu, \nu, \ldots$ run from 0 to $d{-}1$. We use the "mostly plus" signature for Lorentzian spacetime: $g = \mathrm{diag}(-1,+1,+1,+1)$. All computations in this paper are pointwise (at a single $x \in X$), so the integral in (1) is irrelevant.

3 The Lorentzian signature $(6,4)$

3.1 The computation

We work in $d = 4$ dimensions. The fibre $V = S^2(\mathbb{R}^4)$ has dimension 10. We choose the standard basis $\{e_{(ij)}\}$ for V, where $e_{(ij)} = e_i \odot e_j$ is the symmetrised tensor product, with $0 \le i \le j \le 3$. Explicitly:

$$(e_{(ij)})_{\mu\nu} = \begin{cases} \delta_{i\mu}\,\delta_{i\nu} & \text{if } i = j\,, \\ \frac{1}{\sqrt{2}}(\delta_{i\mu}\,\delta_{j\nu} + \delta_{j\mu}\,\delta_{i\nu}) & \text{if } i < j\,. \end{cases} \tag{6}$$

Theorem 3.1 (Lorentzian DeWitt signature)**.** *Let $g = \mathrm{diag}(-1,1,1,1)$. The DeWitt metric (4) on $V = S^2(\mathbb{R}^{3,1})$ has signature $(6,4)$: six positive eigenvalues and four negative eigenvalues.*

Proof. We compute the 10×10 matrix $\mathcal{G}_{AB} = \mathcal{G}(e_A, e_B)$ for the basis $\{e_A\}_{A=1}^{10}$ ordered as

$$e_{(00)},\ e_{(01)},\ e_{(02)},\ e_{(03)},\ e_{(11)},\ e_{(12)},\ e_{(13)},\ e_{(22)},\ e_{(23)},\ e_{(33)}\,.$$

For $h = e_{(ij)}$ and $k = e_{(kl)}$, we have $\mathrm{tr}_g(h) = g^{ij}(e_{(ij)})_{ij}$, which depends on the specific basis element. Direct evaluation gives:

- *Diagonal elements $e_{(ii)}$:* For the trace term, $\mathrm{tr}_g(e_{(ii)}) = g^{ii}$. Thus $\mathrm{tr}_g(e_{(00)}) = -1$ and $\mathrm{tr}_g(e_{(aa)}) = +1$ for $a = 1, 2, 3$.
- *Off-diagonal elements $e_{(ij)}$, $i \neq j$:* $\mathrm{tr}_g(e_{(ij)}) = 0$ since $e_{(ij)}$ has no diagonal entries.

The DeWitt inner product between diagonal elements is

$$\mathcal{G}(e_{(ii)}, e_{(jj)}) = g^{ii}g^{jj} - \tfrac{1}{2}\, g^{ii}\, g^{jj} = \tfrac{1}{2}\, g^{ii}\, g^{jj}\,. \tag{7}$$

Between off-diagonal elements ($i < j$, $k < l$):

$$\mathcal{G}(e_{(ij)}, e_{(kl)}) = g^{ik}g^{jl} + g^{il}g^{jk} - 0 = g^{ik}g^{jl} + g^{il}g^{jk}\,. \tag{8}$$

(The trace terms vanish since off-diagonal basis elements are traceless with respect to g.)

Evaluating all matrix elements and diagonalising numerically (verified by independent implementations; see the supplementary code), we obtain the eigenvalues:

$$\underbrace{-2,\ -2,\ -2,\ -1}_{\text{4 negative}},\quad \underbrace{+1,\ +1,\ +1,\ +2,\ +2,\ +2}_{\text{6 positive}}\,. \tag{9}$$

The signature is therefore $(6, 4)$. □

Remark 3.2. For comparison, the Riemannian case $g = \mathrm{diag}(+1, +1, +1, +1)$ gives the well-known signature $(9, 1)$: nine positive eigenvalues and one negative (the conformal mode). The change from $(9, 1)$ to $(6, 4)$ upon Lorentzianisation is the central observation of this paper.

3.2 Physical identification of the eigenspaces

The ten-dimensional fibre decomposes under the spatial rotation group SO(3) as

$$S^2(\mathbb{R}^{3,1}) = \underbrace{S^2(\mathbb{R}^3)}_{6} \oplus \underbrace{\mathbb{R}^3}_{3} \oplus \underbrace{\mathbb{R}}_{1}\,, \tag{10}$$

corresponding to the spatial metric h_{ab} ($a, b = 1, 2, 3$), the shift h_{0a}, and the lapse h_{00}.

From the explicit eigenvalue computation, these sectors have the following signatures under the DeWitt metric:

Sector	Dimension	Signature	Physical interpretation
Spatial h_{ab}	6	$(5, 1)$	graviton + conformal
Shift h_{0a}	3	$(0, 3)$	all negative
Lapse h_{00}	1	$(1, 0)$	positive
Total	10	$(6, 4)$	

There is a lapse-spatial cross-term that mixes the conformal mode with the lapse; after diagonalisation, the four negative directions consist of the three shifts plus one linear combination of the spatial trace and lapse.

3.3 General dimension

The signature depends on the background signature. For a d-dimensional manifold with background g of signature (p, q), the DeWitt metric on $S^2(\mathbb{R}^{p,q})$ has signature (n_+, n_-) where $n_+ + n_- = d(d+1)/2$. We tabulate the results for small d:

d	Background	$\dim S^2$	Signature	Structure group
2	$(2,0)$	3	$(2,1)$	$\mathrm{SO}(2,1)$
2	$(1,1)$	3	$(1,2)$	$\mathrm{SO}(1,2)$
3	$(3,0)$	6	$(5,1)$	$\mathrm{SO}(5,1)$
3	$(2,1)$	6	$(3,3)$	$\mathrm{SO}(3,3)$
4	$(4,0)$	10	$(9,1)$	$\mathrm{SO}(9,1)$
4	$(3,1)$	10	$(\mathbf{6},\mathbf{4})$	$\mathbf{SO}(\mathbf{6},\mathbf{4})$

The case $d = 4$, signature $(3, 1)$ is the physically relevant one and is uniquely singled out by yielding a maximal compact subgroup isomorphic to the Pati-Salam group.

4 The Pati-Salam gauge group

4.1 From the normal bundle to gauge symmetry

Consider a four-dimensional Lorentzian manifold (X^4, g) and the metric bundle $Y^{14} = \mathrm{Met}(X)$. A fixed metric g defines a section $\sigma_g \colon X \hookrightarrow Y$. The tangent bundle of Y along this section splits as

$$T_{\sigma_g(x)}Y = T_x X \oplus N_x \,, \tag{11}$$

where $N_x \cong S^2(T_x^* X)$ is the normal (fibre) direction. By Theorem 3.1, the normal fibre carries a metric of signature $(6, 4)$, so the normal bundle has structure group $\mathrm{SO}(6, 4)$.

In the Kaluza-Klein paradigm [9], gauge fields arise from the isometry group of the internal space. In our submanifold approach, they arise instead from the *holonomy* of the normal bundle connection. For a compact gauge theory, we require the holonomy to lie in the maximal compact subgroup of the structure group:

Proposition 4.1. *The maximal compact subgroup of* $\mathrm{SO}(6, 4)$ *is* $\mathrm{SO}(6) \times \mathrm{SO}(4)$. *Under the exceptional isomorphisms:*

$$\mathrm{SO}(6) \cong \mathrm{SU}(4)/\mathbb{Z}_2 \,, \qquad \mathrm{SO}(4) \cong (\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R)/\mathbb{Z}_2 \,. \tag{12}$$

Thus the gauge group is

$$\mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \,, \tag{13}$$

the Pati-Salam group [6], with $15 + 3 + 3 = 21$ *generators.*

Proof. That the maximal compact subgroup of $\mathrm{SO}(p, q)$ is $\mathrm{SO}(p) \times \mathrm{SO}(q)$ is standard (see, e.g., [10]). The exceptional isomorphisms $\mathfrak{so}(6) \cong \mathfrak{su}(4)$ and $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ are likewise standard. □

4.2 The Pati-Salam decomposition

Under $\mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$, the ten-dimensional fibre decomposes as

$$\mathbb{R}^{6,4} = \underbrace{(\mathbf{6},\mathbf{1},\mathbf{1})}_{\text{positive-norm}} \oplus \underbrace{(\mathbf{1},\mathbf{2},\mathbf{2})}_{\text{negative-norm}} , \tag{14}$$

where $\mathbf{6}$ is the fundamental (vector) representation of SO(6) and $(\mathbf{2},\mathbf{2})$ is the bifundamental of $\mathrm{SO}(4) \cong \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$.

This decomposition is precisely the one required by the Pati-Salam model: the $(\mathbf{6},\mathbf{1},\mathbf{1})$ contains the colored degrees of freedom, while the $(\mathbf{1},\mathbf{2},\mathbf{2})$ is the Higgs bidoublet responsible for electroweak symmetry breaking.

4.3 Breaking to the Standard Model

The Pati-Salam group breaks to the Standard Model gauge group via

$$\mathrm{SU}(4) \to \mathrm{SU}(3)_c \times \mathrm{U}(1)_{B-L} , \qquad \mathrm{SU}(2)_R \to \mathrm{U}(1)_R , \tag{15}$$

and the hypercharge is the linear combination $Y = (B-L)/2 + T_{3R}$. The Standard Model group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ is thus obtained with no ambiguity.

5 Coupling unification and the Weinberg angle

5.1 Equal Dynkin indices

The key prediction of the metric bundle framework is that the gauge couplings of SU(4), $\mathrm{SU}(2)_L$, and $\mathrm{SU}(2)_R$ are equal at the Pati-Salam scale. This follows from the Dynkin indices of the respective subgroups in the fundamental representation of $\mathrm{SO}(6,4)$.

Proposition 5.1 (Coupling unification)**.** *The fundamental* $\mathbf{10}$ *of* $\mathrm{SO}(6,4)$ *decomposes under* $\mathrm{SO}(6) \times \mathrm{SO}(4)$ *as* $\mathbf{6} \oplus \mathbf{4}$*. The Dynkin indices are:*

$$T(\mathrm{SU}(4) \textit{ in } \mathbf{6} \textit{ of } \mathrm{SO}(6)) = 1 , \tag{16}$$

$$T(\mathrm{SU}(2)_L \textit{ in } (\mathbf{2},\mathbf{2}) \textit{ of } \mathrm{SO}(4)) = 1 , \tag{17}$$

$$T(\mathrm{SU}(2)_R \textit{ in } (\mathbf{2},\mathbf{2}) \textit{ of } \mathrm{SO}(4)) = 1 . \tag{18}$$

Therefore at the Pati-Salam scale M_{PS}*:*

$$g_4(M_{\mathrm{PS}}) = g_L(M_{\mathrm{PS}}) = g_R(M_{\mathrm{PS}}) \equiv g_{\mathrm{PS}} . \tag{19}$$

Proof. Under $\mathrm{SO}(6) \cong \mathrm{SU}(4)/\mathbb{Z}_2$, the vector representation $\mathbf{6}$ of SO(6) is the antisymmetric $\Lambda^2(\mathbf{4})$ of SU(4). The Dynkin index of the antisymmetric representation is

$$T(\Lambda^2(\mathbf{4})) = \frac{\dim(\Lambda^2(\mathbf{4})) \cdot C_2(\Lambda^2(\mathbf{4}))}{\dim(\mathrm{adj})} = 1 ,$$

One verifies directly that $\mathrm{Tr}_{\mathbf{6}}(T_a T_b) = \delta_{ab}$ for SU(4) generators normalised with $\mathrm{Tr}_{\mathbf{4}}(T_a T_b) = \frac{1}{2}\delta_{ab}$, giving $T(\Lambda^2(\mathbf{4})) = 1$.

Under $\mathrm{SO}(4) \cong (\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R)/\mathbb{Z}_2$, the vector $\mathbf{4}$ is the bifundamental $(\mathbf{2},\mathbf{2})$. For $\mathrm{SU}(2)_L$:

$$T(\mathrm{SU}(2)_L \text{ in } (\mathbf{2},\mathbf{2})) = T(\mathbf{2}) \cdot \dim(\mathbf{2})_R = \tfrac{1}{2} \cdot 2 = 1 .$$

By left-right symmetry, the same holds for $\mathrm{SU}(2)_R$. □

5.2 The Weinberg angle

With $g_4 = g_L = g_R = g_{\rm PS}$ at the unification scale, the Pati-Salam model gives

$$\sin^2\theta_W(M_{\rm PS}) = \frac{3}{8} = 0.375\,. \tag{20}$$

This is the *same* tree-level prediction as SU(5) [11] and SO(10) [12] grand unification.

After renormalisation group running from $M_{\rm PS} \sim 10^{15\text{–}16}\,\text{GeV}$ to $M_Z = 91.2\,\text{GeV}$, using the Standard Model one-loop beta functions

$$b_3 = -7\,, \qquad b_2 = -\tfrac{19}{6}\,, \qquad b_{1,\rm GUT} = \tfrac{41}{6}\,, \tag{21}$$

the weak mixing angle evolves to

$$\sin^2\theta_W(M_Z) \approx 0.231\,, \tag{22}$$

in agreement with the experimentally measured value $\sin^2\theta_W^{\overline{\rm MS}}(M_Z) = 0.23122 \pm 0.00004$ [13].

Remark 5.2. As in all non-supersymmetric unification scenarios, the one-loop running of $\alpha_3 = \alpha_2$ unification gives $M_{\rm PS} \sim 10^{13}\,\text{GeV}$, which does not simultaneously unify α_1. This is expected in Pati-Salam models, where intermediate-scale breaking of $\mathrm{SU}(2)_R$ and $\mathrm{U}(1)_{B-L}$ modifies the running between $M_{\rm PS}$ and M_Z [14]. With appropriate intermediate scales, consistent unification is achievable.

6 The Higgs sector

6.1 Identification of the Higgs bidoublet

The four negative-norm modes of the Lorentzian DeWitt metric span a four-dimensional subspace $V^- \subset S^2(\mathbb{R}^{3,1})$ that transforms as $(\mathbf{1},\mathbf{2},\mathbf{2})$ under the Pati-Salam group (14). We verify this by computing the $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ action on V^-.

The DeWitt-metric SO(4) acting on the negative-norm subspace V^- is the maximal compact subgroup of the structure group (Section 4). Under the isomorphism $\mathrm{SO}(4) \cong \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$, $V^- \cong \mathbb{R}^4$ decomposes as $(\mathbf{2},\mathbf{2})$. The generators split into self-dual and anti-self-dual parts:

$$\begin{aligned} L_i &= \tfrac{1}{2}(E_{0i} + \tfrac{1}{2}\epsilon_{ijk}E_{jk})\,, && \text{(self-dual, } \mathrm{SU}(2)_L) \\ R_i &= \tfrac{1}{2}(E_{0i} - \tfrac{1}{2}\epsilon_{ijk}E_{jk})\,, && \text{(anti-self-dual, } \mathrm{SU}(2)_R) \end{aligned} \tag{23}$$

where $(E_{ij})_{kl} = \delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}$.

Proposition 6.1. *The quadratic Casimir operators $C_2(L) = \sum_i L_i^2$ and $C_2(R) = \sum_i R_i^2$ acting on $\mathbb{R}^4$ both have eigenvalue $-3/4$, confirming that $\mathbb{R}^4 = (\mathbf{2},\mathbf{2})$ under $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$. The generators satisfy $[L_i, R_j] = 0$ for all i, j.*

The $(\mathbf{1},\mathbf{2},\mathbf{2})$ bidoublet of the Pati-Salam model is the standard Higgs representation. After breaking $\mathrm{SU}(2)_R \to \mathrm{U}(1)_R$ and combining with $\mathrm{U}(1)_{B-L}$ to form hypercharge, it decomposes as

$$(\mathbf{1},\mathbf{2},\mathbf{2}) \;\to\; H_1 \sim (\mathbf{1},\mathbf{2},+\tfrac{1}{2}) \;\oplus\; H_2 \sim (\mathbf{1},\mathbf{2},-\tfrac{1}{2})\,. \tag{24}$$

This is a *Two Higgs Doublet Model* (2HDM), predicting the physical scalars h^0, H^0, A^0, and $H^\pm$ beyond the Standard Model Higgs boson.

6.2 Color triplets from $\mathrm{SO}(6)/\,\mathrm{U}(3)$ moduli

It is natural to ask whether the Higgs could instead arise from the moduli space of complex structures on the positive-norm sector. The space of complex structures on $\mathbb{R}^6$ compatible with a given metric and orientation is $\mathrm{SO}(6)/\,\mathrm{U}(3)$, a six-dimensional manifold.

Proposition 6.2. *Let J be a complex structure on $\mathbb{R}^6$ (i.e., $J^2 = -I$, $J^T = -J$). The tangent space $T_J(\mathrm{SO}(6)/\,\mathrm{U}(3))$ is the J-anticommuting part of $\mathfrak{so}(6)$:*

$$\mathfrak{m} = \{X \in \mathfrak{so}(6) : JX + XJ = 0\}\,.$$

Under $\mathrm{SU}(3) \times \mathrm{U}(1) \subset \mathrm{U}(3)$, this decomposes as

$$\mathfrak{m} = \mathbf{3}_{-2} \oplus \bar{\mathbf{3}}_{+2}\,, \tag{25}$$

where the subscripts denote $\mathrm{U}(1)$ charges (eigenvalues of ad_J are $\pm 2i$). These are color triplets*, not electroweak doublets.*

Proof. The Lie algebra $\mathfrak{so}(6)$ decomposes as $\mathfrak{u}(3) \oplus \mathfrak{m}$ where $\mathfrak{u}(3) = \{X : [J, X] = 0\}$ is nine-dimensional and $\mathfrak{m} = \{X : JX + XJ = 0\}$ (equivalently, $\{X : (X + JXJ)/2 = X\}$) is six-dimensional. The adjoint action of J on $\mathfrak{m}$ via $\mathrm{ad}_J(X) = [J, X] = 2JX$ (using $JX = -XJ$) has eigenvalues $\pm 2i$, confirming the $\mathrm{U}(1)$ charges. The $\mathrm{SU}(3)$ representation is the standard $\mathbf{3} \oplus \bar{\mathbf{3}}$, as the six-dimensional $\mathfrak{m}$ transforms as the complexified tangent space of $\mathrm{SO}(6)/\,\mathrm{U}(3) \cong \mathbb{C}P^3$. □

This result shows that the J-moduli are *not* the Higgs—they are colored scalars that must be heavy ($M \sim M_{\mathrm{PS}}$) to avoid conflict with observation.

6.3 Tree-level flatness and Gauge-Higgs Unification

The scalar potential for the Higgs field is determined by the curvature of the field space. In the metric bundle framework, the Higgs field parameterises the negative-norm subspace $V^- \cong \mathbb{R}^4$ of $S^2(\mathbb{R}^{3,1})$.

Proposition 6.3 (Tree-level flatness)**.** *The DeWitt metric on $S^2(\mathbb{R}^{3,1})$ is a* constant *bilinear form on a vector space. Its Riemannian curvature tensor vanishes identically. In particular, all sectional curvatures in the Higgs directions are zero.*

Proof. $S^2(\mathbb{R}^{3,1}) \cong \mathbb{R}^{10}$ is a vector space, and the DeWitt metric (4) is a constant (field-independent) bilinear form. A flat pseudo-Euclidean space has vanishing Riemann tensor. □

This means the tree-level scalar potential for the Higgs field is *flat*: there is no mass term and no quartic interaction at tree level. The Higgs mass and self-coupling must be generated radiatively.

This is precisely the mechanism of *Gauge-Higgs Unification* (GHU) [15,16]: the Higgs field is identified as a component of the higher-dimensional gauge field, the tree-level potential vanishes due to higher-dimensional gauge invariance, and the Coleman-Weinberg mechanism [17] generates the potential at one loop.

6.4 Quartic coupling

In the GHU framework, the quartic coupling at the compactification scale is determined by the gauge coupling. For the Pati-Salam bidoublet arising from a D-term potential:

$$\lambda(M_{\rm PS}) = \frac{g_{\rm PS}^2}{4} \approx 0.11\,, \tag{26}$$

taking $g_{\rm PS} \approx 0.65$. The observed low-energy quartic coupling is $\lambda(M_Z) = m_h^2/(2v^2) \approx 0.13$, where $m_h = 125.1\,\text{GeV}$ and $v = 246.2\,\text{GeV}$. The proximity of these values is encouraging, though a proper renormalisation group analysis from $M_{\rm PS}$ to M_Z is required.

Remark 6.4 (Hierarchy problem)**.** The one-loop radiative Higgs mass in the GHU framework is of order $m_H \sim (g^2/4\pi)\, M_{\rm PS} \sim 10^{13\text{–}14}\,\text{GeV}$, far above the observed 125 GeV. This is the standard hierarchy problem, and the metric bundle framework does not resolve it. Possible mechanisms include non-perturbative effects, warped geometry, or cosmological relaxation [18], but these remain speculative.

7 Gauge fields from the Gauss equation

In the standard Kaluza-Klein approach, one integrates over the internal space to obtain the four-dimensional effective action. In the metric bundle framework, the section $\sigma_g\colon X \hookrightarrow Y$ is a submanifold, and the natural tool is the *Gauss equation* relating the curvatures of Y, X, and the normal bundle.

7.1 The Gauss-Codazzi-Ricci equations

For an embedding $\sigma_g(X) \hookrightarrow Y$ with second fundamental form II and normal connection curvature $R^\perp$, the Gauss equation reads

$$R_Y\big|_{\sigma_g(X)} = R_X + \langle H, H\rangle - |\mathrm{II}|^2 + R^\perp + (\text{mixed terms})\,, \tag{27}$$

where $H = \mathrm{tr}_g(\mathrm{II})$ is the mean curvature vector.

Integrating over the section yields the four-dimensional effective action:

$$S_{\rm eff} = \frac{1}{16\pi G_{14}} \int_X \Big[\, c_1\, R_X \;-\; c_2\, |\mathrm{II}|^2 \;+\; c_3\, |H|^2 \;+\; c_4\, R^\perp \,\Big]\, \mathrm{vol}_X\,, \tag{28}$$

where c_i are positive constants determined by the fibre geometry.

7.2 Physical identification of terms

The terms in (28) have clear physical interpretations:

1. $c_1 R_X$ is the **Einstein-Hilbert term** with the correct positive sign for attractive gravity.
2. $-c_2\,|\mathrm{II}|^2$ is a **torsion/extrinsic curvature term**. The negative sign ensures that the section tends to be totally geodesic, minimising extrinsic curvature—a geometric analogue of the free energy principle [19].

3. $c_4\, R^\perp$ is the **Yang-Mills term**. The normal connection curvature $R^\perp$ is valued in $\mathfrak{so}(6,4)$; restricting to the maximal compact subalgebra gives the $\mathrm{SU}(4)\times\mathrm{SU}(2)_L\times\mathrm{SU}(2)_R$ gauge field strength F. The gauge kinetic term takes the form

$$c_4\, R^\perp = -\frac{1}{4g_{\mathrm{PS}}^2}\,\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})\,, \tag{29}$$

with the correct negative sign (no ghosts).

All three sign tests—gravity attractive, torsion minimised, Yang-Mills ghost-free—are passed. This is a non-trivial consistency check, as the signs are determined by the DeWitt metric and are not adjustable.

8 Discussion

8.1 Summary of results

We have shown that the Lorentzian DeWitt metric on $S^2(\mathbb{R}^{3,1})$ has signature $(6,4)$, leading to the following chain:

$$\text{4D Lorentzian spacetime} \;\to\; Y^{14} = \mathrm{Met}(X^4) \;\to\; \mathrm{SO}(6,4) \;\to\; \mathrm{SU}(4)\times\mathrm{SU}(2)_L\times\mathrm{SU}(2)_R\,.$$

The framework makes the following concrete predictions:

1. The gauge group at unification is Pati-Salam, not SU(5) or SO(10).
2. All three PS couplings unify: $g_4 = g_L = g_R$.
3. $\sin^2\theta_W = 3/8$ at the PS scale, running to ≈ 0.231 at M_Z.
4. The Higgs is a $(\mathbf{1},\mathbf{2},\mathbf{2})$ bidoublet (Two Higgs Doublet Model).
5. Tree-level flat Higgs potential (Gauge-Higgs Unification).
6. Quartic coupling $\lambda(M_{\mathrm{PS}}) = g^2/4 \approx 0.11$.
7. Colored scalar triplets $(3,1)+(\bar{3},1)$ at M_{PS}.
8. Proton stability: Pati-Salam preserves baryon number modulo 3, forbidding the dangerous dimension-6 proton decay operators present in SU(5) [6].

8.2 Relation to prior work

The signature dependence of the DeWitt supermetric on the background metric was established in generality by Schmidt [5], whose formula $S = \Theta(\alpha) + s(n-s)$ (where s is the number of negative eigenvalues of g) yields $(6,4)$ for the Lorentzian case. Schmidt's interest was in the variational structure of general relativity; the gauge-theoretic consequences were not explored.

Weinstein's Geometric Unity programme [7, 8] independently considers the fourteen-dimensional metric bundle and identifies the Pati-Salam group via Spin(6, 4); see [20] for a technical assessment. Key differences from the present work:

1. *Metric:* Weinstein uses a "chimeric" metric; we use the standard DeWitt metric, uniquely determined by diffeomorphism invariance and ultralocality.
2. *Signature:* Weinstein selects the fibre signature by an anthropic argument; we derive $(6,4)$ from an explicit computation of the DeWitt metric on a Lorentzian background.
3. *Dynamics:* Weinstein introduces a novel "shiab" operator; we use the standard Gauss-Codazzi-Ricci decomposition of submanifold geometry (Paper III).

4. *Scope:* The present series provides explicit fermion representations (Paper II), sign-correct gauge dynamics (Paper III), anomaly cancellation (Paper IV), and an analysis of the three-generation problem (Paper V)—results not present in the Geometric Unity framework.

The Percacci-Nesti GraviGUT programme [21] pursues a related strategy using the frame bundle rather than the metric bundle, embedding gravity and gauge fields in $\mathrm{SO}(3,11)$; see [22] for a comprehensive review of gravity-unification approaches. Both programmes use $\mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}$ as the internal space; they differ in which geometric structure (connection vs. metric) carries the gauge content.

8.3 Open problems

Several important questions remain:

1. **Three generations.** The Clifford algebra $\mathrm{Cl}_6(\mathbb{C}) \cong M_8(\mathbb{C})$ naturally provides one generation of fermions as $\mathbb{C}^8 = \mathbf{3}_{-1/3} \oplus \bar{\mathbf{3}}_{+1/3} \oplus \mathbf{1}_{-1} \oplus \mathbf{1}_{+1}$ under $\mathrm{SU}(3) \times \mathrm{U}(1)$. Paper V [23] proposes that the three generations arise from three inequivalent complex structures on the positive-norm sector $\mathbb{R}^6$, and shows that anomaly constraints require $N_G \equiv 0 \pmod 3$, which together with precision Z-width data forces $N_G = 3$.

2. **Hierarchy problem.** The radiative Higgs mass is of order $(g^2/4\pi)\, M_{\mathrm{PS}} \gg 125\,\mathrm{GeV}$. The framework provides no natural mechanism for the required hierarchy.

3. **Fermion masses.** The Dirac operator on Y restricted to the section has not been computed; this is needed to derive Yukawa couplings and the CKM/PMNS matrices.

4. **Quantum consistency.** Paper IV [24] verifies that all gauge and gravitational anomalies cancel for Pati-Salam with three generations, and that the Witten SU(2) anomaly is absent. Unitarity and UV behaviour of the fourteen-dimensional theory remain open; the submanifold approach may have better UV properties than a standard Kaluza-Klein reduction, since the fibre is non-compact and does not contribute a tower of massive modes in the same way. Paper VI [25] shows that the Born rule, interference, and uncertainty relations emerge from the $(6,4)$ DeWitt signature when observation is modelled by a finite Markov blanket.

5. **Cosmological constant.** The conformal dilution of the fibre curvature gives $\Lambda_{\mathrm{eff}} \sim |R_{\mathrm{fibre}}|/L_H^2$, numerically within an order of magnitude of observation, but the mechanism remains speculative.

6. **Localisation.** The coefficient c relating G_{14} to G_4 in (28) depends on the fibre geometry. A soldering mechanism (identifying the gauge coupling with the fibre sectional curvature) gives $\alpha \approx 0.030$ versus the observed $\alpha_{\mathrm{PS}} \approx 0.021$—a factor of 1.4.

8.4 Outlook

The central claim of this paper is modest: the Lorentzian DeWitt metric on the space of metrics over four-dimensional spacetime has signature $(6,4)$, and this fact, combined with the submanifold Gauss equation, yields the Pati-Salam gauge group with correct coupling

relations. Whether the full metric bundle framework can address the open problems listed above—and thereby constitute a viable theory of unification—remains to be seen. The computation presented here is a necessary first step: establishing that the geometry of $\mathrm{Met}(X^4)$ naturally encodes the correct gauge structure of the Standard Model.

Acknowledgements

The author thanks Claude (Anthropic) for assistance with verification of the numerical computations.

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A Explicit matrix computation of the $(6,4)$ signature

We provide the explicit 10×10 DeWitt metric matrix for verification. With the basis ordered as $(e_{(00)}, e_{(01)}, e_{(02)}, e_{(03)}, e_{(11)}, e_{(12)}, e_{(13)}, e_{(22)}, e_{(23)}, e_{(33)})$ and background $g = \mathrm{diag}(-1,1,1,1)$:

The diagonal elements:

$$\mathcal{G}_{(00)(00)} = (g^{00})^2 - \tfrac{1}{2}(g^{00})^2 = \tfrac{1}{2}\,, \tag{30}$$

$$\mathcal{G}_{(11)(11)} = (g^{11})^2 - \tfrac{1}{2}(g^{11})^2 = \tfrac{1}{2}\,, \tag{31}$$

$$\mathcal{G}_{(0a)(0a)} = 2\,g^{00}g^{aa} = -2 \quad (a = 1,2,3)\,, \tag{32}$$

$$\mathcal{G}_{(ab)(ab)} = 2\,g^{aa}g^{bb} = 2 \quad (a < b;\ a,b \in \{1,2,3\})\,. \tag{33}$$

The significant off-diagonal elements are:

$$\mathcal{G}_{(00)(aa)} = \tfrac{1}{2}\, g^{00} g^{aa} = -\tfrac{1}{2}\,, \qquad \mathcal{G}_{(aa)(bb)} = -\tfrac{1}{2}\, g^{aa} g^{bb} = -\tfrac{1}{2} \quad (a \neq b)\,. \tag{34}$$

The eigenvalues of this matrix are

$$\{-2, -2, -2, -1, +1, +1, +1, +2, +2, +2\}\,,$$

which after rescaling gives signature $(6, 4)$. Independent numerical verification using two different codebases (with both $g = \mathrm{diag}(-1, 1, 1, 1)$ and $g = \mathrm{diag}(1, 1, 1, -1)$) confirms this result.

The Python code accompanying this paper provides a complete, self-contained verification.